

QUESTIONS

These released questions are from the mathematics component of the junior-division assessment. This section provides the overall expectation and the category of knowledge and skills for each question.

B. NUMBER

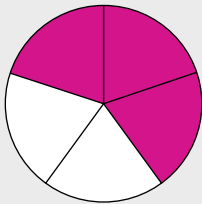
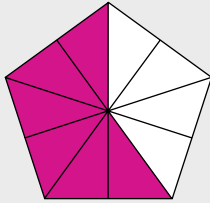
B1. Number Sense

demonstrate an understanding of numbers and make connections to the way numbers are used in everyday life

1 Which number is between 2.354 and 2.456?
KU

- A 2.345
- B 2.402
- C 2.457
- D 2.520

2 Select the option that is **not** equal to 6 tenths.
AP

- A 
- B $\frac{6}{10}$
- C 0.06
- D 

B2. Operations

use knowledge of numbers and operations to solve mathematical problems encountered in everyday life

3 Select the **TWO** prime numbers.

KU

A 11

B 21

C 24

D 42

E 43

4 A jar contains 200 marbles, and 5% of the marbles are blue.

AP

How many marbles are blue?

A 5 marbles

B 10 marbles

C 20 marbles

D 25 marbles

B2. Operations (continued)

5 Lucas types 360 words in 12 minutes.

TH

Greg types twice as fast as Lucas.

How many words per minute does Greg type?

A 15 words per minute

B 30 words per minute

C 60 words per minute

D 180 words per minute

C. ALGEBRA

C1. Patterns and Relationships

identify, describe, extend, create, and make predictions about a variety of patterns, including those found in real-life contexts

6 Four term values of a growing pattern are shown.
AP

Term number	5	7	9	
Term value	11	15	19	27

What is the term number when the term value is 27?

- A

11
- B

13
- C

15
- D

23

C1. Patterns and Relationships (continued)

7 This table shows a starting number and a rule for three patterns.
TH

	Pattern P	Pattern Q	Pattern R
Starting number	1	66	4
Rule	Add 5	Subtract 10	Multiply by 2

Which number is in all three patterns?

- A

8
- B

16
- C

26
- D

36

C2. Equations and Inequalities

demonstrate an understanding of variables, expressions, equalities, and inequalities, and apply this understanding in various contexts

8 What value of x makes this equation true?

KU

$$7x = 28$$

A 3

B 4

C 21

D 35

9 In which equation does p have the greatest value?

TH

A $85 + p = 125$

B $70 - p = 35$

C $5p = 215$

D $3p = 132$

D. DATA

D1. Data Literacy

manage, analyse, and use data to make convincing arguments and informed decisions, in various contexts drawn from real life

11 **KU** This table shows the number of bags of popcorn sold at an event during different times in a day.

Time of day	Number of bags of popcorn sold
1:00–2:59	120
3:00–4:59	87
5:00–6:59	113

Which option would **best** represent this data?

- A histogram
- B line plot
- C pictograph
- D multiple-bar graph

D1. Data Literacy (continued)

12 The height in centimetres of various plants is shown.
TH

16	21	21	31	32	34	41
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Drag and drop the numbers that complete each statement about this set of numbers.

16

21

25

28

31

41

The range is cm.

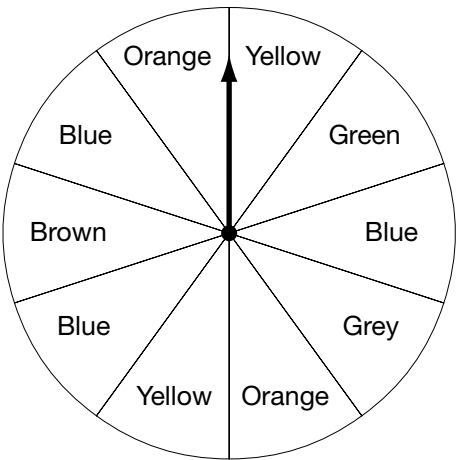
The median is cm.

The mean is cm.

D2. Probability

describe the likelihood that events will happen, and use that information to make predictions

13 This spinner is divided into 10 equal sections.
AP



The arrow on the spinner is spun.
What is the probability that the arrow will land on yellow?

- A 10%
- B 20%
- C 40%
- D 80%

E. SPATIAL SENSE

E1. Geometric and Spatial Reasoning

describe and represent shape, location, and movement by applying geometric properties and spatial relationships in order to navigate the world around them

14 Which of these shapes has exactly two lines of symmetry?
KU

A



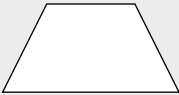
Square

B



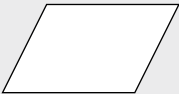
Rhombus

C



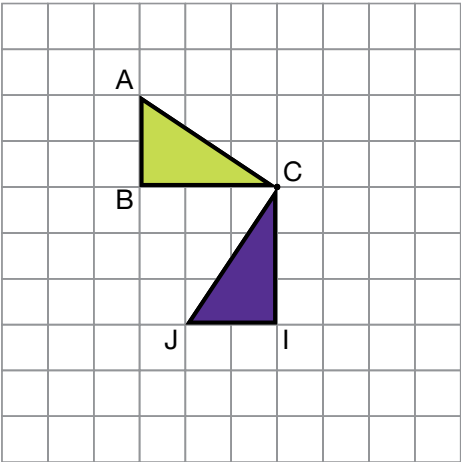
Trapezoid

D



Parallelogram

15 Which rotation about point C moves $\triangle ABC$ onto $\triangle CIJ$?
AP



A

90° clockwise

B

90° counterclockwise

C

180° clockwise

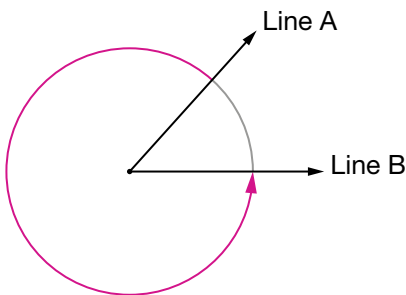
D

180° counterclockwise

E2. Measurement

compare, estimate, and determine measurements in various contexts

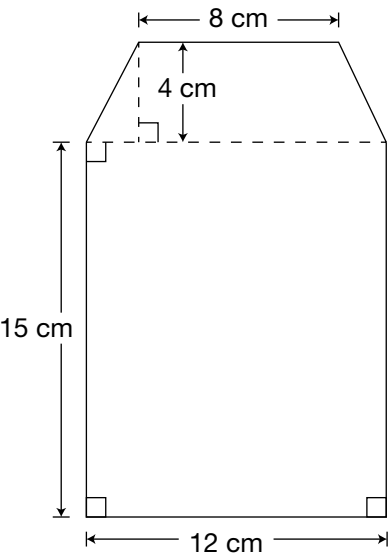
- 16** The acute angle between Line A and Line B is 48° .
AP



What is the angle when measuring **counterclockwise** from Line A to Line B?

- A
- B
- C
- D

- 17** What is the **total** area of this polygon?
TH



- A
- B
- C
- D